

Training and evaluating the model

Nested cross-validation for CatBoost · 187 patients · random seed 42

■ used to fit ■ inner validation ■ outer test (held out) ■ search / decision

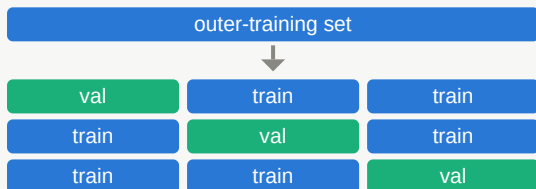
Which patients are held out

Repeated Stratified K-Fold: 4 folds × 10 repeats = 40 outer iterations



Which training patients tune the model

Stratified K-Fold: 3-fold, outer-training set only



Optuna TPE — 100 trials per outer iteration

Objective: mean AUPRC across the 3 validation folds

Tunes: depth · learning rate · L2 leaf reg · pos-class weight

↓ carried forward: winning hyperparameters + the same 3 inner splits

Refit, threshold, and outer-test evaluation

· per iteration

1 · Refit

Winning params refit on the full outer-training set

2 · OOF scores

One score per patient

3 · Threshold

Maximize F1 on OOF curve

4 · Outer-test evaluation

Model + threshold scored once
→ performance metrics and plots